

Linear Relaxation and Lagrangian Relaxation

Linear Relaxation and HiGHS Solver

In the current Lagrangian relaxation approach, the demand coverage constraint is dualized while the binary maintenance variables remain unchanged. Consequently, the subproblem $P2$ remains a MILP with most of the original constraints.

Linear relaxation removes the integrality requirements on the variables :

$$x_{itk}, y_{it} \in \{0, 1\} \rightarrow x_{itk}, y_{it} \in [0, 1]$$

Lagrangian relaxation allows for the relaxation of coupling constraints by displacing them in the objective function with Lagrange multipliers.

When solving a MILP, HiGHS internally solves the LP (Linear Programming) relaxation as part of its branch-and-bound procedure. To obtain bounds and dual information, the solver begins by relaxing binary variables and solving the resulting LP. It then applies branch-and-bound to recover a feasible integer solution. The $P2$ MILP subproblem can be solved without completing a separate LP relaxation because HiGHS automatically performs this step.

By applying LP relaxation directly to the Lagrangian subproblem $P2$, both approaches can be combined. In this case, $P2$ becomes a linear program instead of a MILP. This generally leads to faster resolution times, but can produce weaker dual bounds.

Method	$P2$	Speed	Dual bound
Current Lagrangian relaxation	MILP	Slower	Stronger
Lagrangian + LP relaxation	LP	Faster	Usually weaker

TABLE 1 – Trade-off between the two Lagrangian approaches.

If the binary variables are relaxed in $P2$, no additional constraints are needed for the model to remain mathematically valid. All original constraints still apply because only the variable domains are changed.

However, the interpretation of the solution changes. The variables y_{it} and x_{itk} may take fractional values, which can represent partially scheduled maintenance or fractional maintenance starts. These solutions are not physically feasible, but they are acceptable within an LP relaxation because the main goal is to obtain bounds or speed up the dual optimization process.

Additional constraints would only be required if the objective were to strengthen the LP relaxation through tighter formulations or valid inequalities. In this work, we first focus on the simple combination of linear and Lagrangian relaxation.

Solution extraction : re-project into a feasible schedule

A re-projection step is therefore required to recover a feasible solution for the original problem. The objective of this step is to transform the fractional solution into a valid maintenance schedule while preserving as much information as possible from the relaxed solution.

There are two approaches to recover an integer and feasible solution from the relaxed schedule.

The first approach is an exact recovery method based on branch-and-bound. In this case, the fractional solution obtained from the relaxed problem is used as an initial solution for a MILP solver. This approach guarantees a feasible integer solution and may improve solution quality, but it can require significant computational time.

The second approach is a heuristic recovery method based on projection or rounding. The fractional maintenance variables are converted directly into binary values using simple decision rules, such as selecting the largest maintenance-start variables or applying threshold rounding. Once the maintenance schedule is fixed, the remaining production and fuel management variables are re-optimized.